# Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

## Lesson and revision-note context

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|---|---|
| **Subject** | Mathematics |
| **Grade** | JSS1 |
| **Term** | Third Term |
| **Week** | 4 |
| **Topic** | Three Dimensional Shapes |
| **Subtopic** | Three dimensional figures: properties of cubes, cuboids, pyramids and cones |
| **Theme** | — |
| **Curriculum objectives** | • Identify the properties of cube and cuboids<br>• Identify the properties of pyramids and cones |
| **Previous knowledge** | • Students have been taught the differences between cube and cuboid. |
| **Lesson sequence** | • Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks<br>• Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II |
| **Instructional materials** | • Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres. |
| **Evaluation** | • How many vertices, has a cuboid?<br>• How many edges has a cube?<br>• Draw a cube and cuboid then label.<br>• Draw an hexagonal base pyramid and list the properties.<br>• How many vertices has a tetrahedron. |

## Lesson objectives

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|---|---|
| **1.** | Identify the properties of cube and cuboids |
| **2.** | Identify the properties of pyramids and cones |

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

# STUDENT REVISION NOTE

**Topic:** Three dimensional figures: properties of cubes, cuboids, pyramids and cones

**Class:** JSS1 **Subject:** Mathematics **Week:** 4

## Section 1: Cubes and Cuboids: Faces, Edges and Vertices

When we talk about a solid shape like a cube or a cuboid, we describe it by counting its faces, edges and vertices.

- A face is a flat surface of the shape.
- An edge is the line where two faces meet.
- A vertex (plural: vertices) is a corner where edges meet.

<h3>The cube</h3>
A cube is a solid in which every face is a square of the same size.

- Faces: 6 (all squares, and they are all equal).
- Edges: 12 (and they are all the same length).
- Vertices: 8.

Because every edge of a cube is the same length, the cube is the same width, depth and height.

<h3>The cuboid</h3>
A cuboid is a solid whose faces are rectangles. It has the same counts as a cube, but its edges are not all the same length.

- Faces: 6 (rectangles). Opposite faces are equal, so the six faces come in three equal pairs.
- Edges: 12. They come in three groups of four, and the four edges in each group are equal.
- Vertices: 8.

A cuboid has a length, a width and a height, and those three measurements are usually different. If you imagine making all three equal, the cuboid becomes a cube — that is why a cube is a special kind of cuboid.

<h3>Worked example — labelling a cuboid</h3>
Here is how you check and label a cuboid before you answer any question about it.

Imagine the box of a shoe. Its top, its base and its four sides are all rectangles. Counting them:

- The top is one face and the base is another face, so that is 2.
- The four sides are 4 more faces.
- Total faces = 6. ✓

Now look at one corner of the box: three edges meet there. Count the edges around the box — top has 4, base has 4, and the four vertical edges join them — that gives 4 + 4 + 4 = 12 edges. ✓

Finally, each corner of the box is a vertex. There are 4 corners on top and 4 on the bottom, so the box has 8 vertices. ✓

So a cuboid always has 6 faces, 12 edges and 8 vertices, and its opposite faces are equal.

**Practice**

1. How many vertices does a cuboid have?
2. How many edges does a cube have?
3. A solid shape has 6 faces, 12 edges and 8 vertices. Its faces are all equal squares. What is the shape called, and how long is each edge if the shape is 4 cm wide, 4 cm deep and 4 cm high?

## Section 2: Pyramids and Cones: Getting to Know Their Properties

In this section you will learn to identify the properties of pyramids and cones. Just as you did for cubes and cuboids, you will meet each shape, learn what it is made of, and list its properties — the number of faces, edges and vertices, and the shapes of those faces.

A pyramid is a three-dimensional shape with a flat base that is a polygon (a shape with straight sides) and triangular faces that all meet at a single point at the top, called the apex. The name of a pyramid tells you the shape of its base.

Let's look at two pyramids.

A square-based pyramid has a square base. Its properties are:
- It has 5 faces: 1 square base and 4 triangular faces.
- It has 8 edges.
- It has 5 vertices.

A triangular-based pyramid has a triangular base. Its properties are:
- It has 4 faces: 1 triangular base and 3 triangular faces.
- It has 6 edges.
- It has 4 vertices.

A triangular-based pyramid is also called a tetrahedron. Notice its name: "tetra" means four, and it has 4 faces. When someone asks how many vertices a tetrahedron has, the answer is 4.

A cone is a three-dimensional shape with a flat circular base and a curved surface that narrows to a single point, called the apex. Its properties are:
- It has 2 faces: 1 flat circular base and 1 curved surface.
- It has 1 vertex (the point at the top).
- It has no edges, because the base is a circle and the curved surface has no straight edge.

So when you list the properties of a cone, remember it has one curved surface and one flat circular base, one vertex at its apex, and no edges.

The key difference between a pyramid and a cone is the base: a pyramid has a polygonal (straight-sided) base, while a cone has a circular base.

Let's work through an example together: an hexagonal base pyramid.

An hexagonal base pyramid has a base that is a hexagon (a six-sided polygon), and each side of the hexagon has one triangular face rising to the apex. So:

- Faces: 1 hexagonal base + 6 triangular faces = 7 faces.
- Edges: 6 edges around the hexagonal base + 6 edges from the base up to the apex = 12 edges.
- Vertices: the 6 vertices around the base + 1 apex = 7 vertices.

So an hexagonal base pyramid has 7 faces, 12 edges and 7 vertices.

Now try these for yourself.

**Practice**

1. List the properties (number of faces, edges and vertices) of a square-based pyramid.
2. How many vertices does a tetrahedron have?
3. List the properties (number of faces, edges and vertices) of a cone.
4. Draw an hexagonal base pyramid and list its properties.

## Check your work

Attempt every question before you read this page.

### Cubes and Cuboids: Faces, Edges and Vertices

1. A cuboid has 8 vertices.
2. A cube has 12 edges.
3. The shape is a cube, and each edge is 4 cm.

### Pyramids and Cones: Getting to Know Their Properties

1. A square-based pyramid has 5 faces (1 square base and 4 triangular faces), 8 edges and 5 vertices.

Count the faces: 1 square base plus 4 triangular faces = 5 faces. Count the edges: 4 around the square base plus 4 from the base up to the apex = 8 edges. Count the vertices: the 4 corners of the square base plus the 1 apex = 5 vertices.

2. A tetrahedron (a triangular-based pyramid) has 4 vertices.

   A tetrahedron has a triangular base (3 vertices) plus one apex, so 3 + 1 = 4 vertices.

3. A cone has 2 faces (1 flat circular base and 1 curved surface), 1 vertex and no edges.

   A cone has one flat circular base and one curved surface, giving 2 faces. Its curved surface narrows to a single point, the apex, which is its 1 vertex. Because the base is a circle and there is no straight edge, it has 0 edges.

4. An hexagonal base pyramid has 7 faces (1 hexagonal base and 6 triangular faces), 12 edges and 7 vertices.

   Faces: 1 hexagonal base + 6 triangular faces = 7 faces. Edges: 6 around the hexagonal base + 6 from the base to the apex = 12 edges. Vertices: 6 corners of the base + 1 apex = 7 vertices.

## Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

| | | | |
|---|---|---|---|
| **Revision note ID** | cand-09b52e1b | **Date** | |
| **Subtopic** | Three dimensional figures: properties of cubes, cuboids, pyramids and cones | **Rater** | |

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

| **Criterion** (select one score per row) | **1** Poor | **2** Weak | **3** Adequate | **4** Good | **5** Excellent |
|---|---|---|---|---|---|
| **Objective and curriculum alignment** Does the revision note cover the intended lesson objectives and curriculum target? | ○ | ○ | ○ | ○ | ○ |
| **Taught-lesson alignment** Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth? | ○ | ○ | ○ | ○ | ○ |
| **Scope and conceptual correctness** Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors? | ○ | ○ | ○ | ○ | ○ |
| **Student-facing clarity and usefulness** Can JSS1 learners understand and use it meaningfully as take-home review material? | ○ | ○ | ○ | ○ | ○ |

| Strengths | Concerns | Recommendation |
|---|---|---|
| | | |

## Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

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|---|---|
| ☐ | **Misses a required objective** — Skips something the lesson was meant to teach. |
| ☐ | **Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic). |
| ☐ | **Wrong or misleading content** — Contains a mathematical or factual mistake. |
| ☐ | **Too hard for JSS1 to follow** — Likely confusing for the students. |
| ☐ | **Other:** _____ |